PAPER_1491_NI62_PEAK_BINDING

Daniel T. Murphy

PAPER_1491 — UQFF: Ni-62 Peak Nucleon Binding (extends PAPER_872) (Bucket G)

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket G — derived from F:/Aetheric Propulsion source material
Date: June 16, 2026
Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)
Status: CLOSED — Z = D_crit+2 = 28, N = D_crit+2·D_phys = 34, A = A_5+2 = 62 — all EXACT

Observation

Nickel-62 has the highest binding energy per nucleon in nature (BE/A = 8.7945 MeV, surpassing Fe-56 at 8.790 MeV).

UQFF Closed Identity

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Z(Ni-62) = D_crit + 2 = 28 EXACT; N(Ni-62) = D_crit + 2·D_phys = 34 EXACT;
A(Ni-62) = A_5 + 2 = 62 EXACT.
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Physical Interpretation

Ni-62 extends the proto-element chain established in PAPER_872 (proto-Fe Z = D_crit = 26 EXACT; proto-Si Z = SO_5 + D_phys = 14 EXACT). The +2 offset from D_crit reflects the additional pairing energy above the proto-Fe shell closure. All three nuclear quantum numbers (Z, N, A) reduce to canonical integer-primitive identities.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives {D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60}. Zero free parameters.

Reference

- Source: F:/Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
- Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_872 (proto-element nuclear identity), PAPER_087 (BH thermodynamics), PAPER_1156 (cosmology suite), PAPER_1255 (muonic hydrogen radius).
